# Spatial Noise and the Standard Errors of Persistence

ECON 895 Spatial Techniques in Empirical Economics

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# Plan for tonight

Two persistence results

The standard errors of persistence

Running it with Claude (optional)

Paper clinic

Exercise 2 debrief

We break around 8:30. The second hour is the paper clinic. Claude Code does the computing tonight in a copy of `examples/persistence_project/`.

# Exercise 2

- Exercise 2 was due by email at 7:20 tonight.

# Two persistence results

# Two theories of where cities are

- **Locational fundamentals.** A harbor, a river crossing, fertile land. The map of cities is stable because the geography underneath it is stable.

- **Increasing returns.** People and firms are productive near other people and firms. The map of cities is stable because being large is self-reinforcing, wherever the city happens to be.

- The two theories agree about a stable map. They disagree about a shock. Take the people away and fundamentals say they come back. Take the fundamental away and increasing returns say the people stay.

- Davis and Weinstein (2002) study a shock to the people and Bleakley and Lin (2012) a shock to the fundamental.

# Cities that lost the most people during the war grew fastest after it

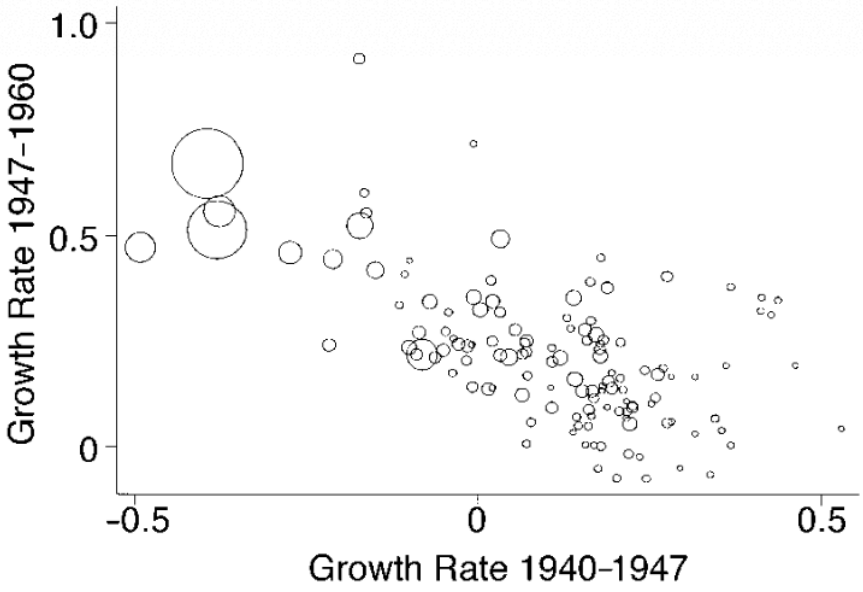
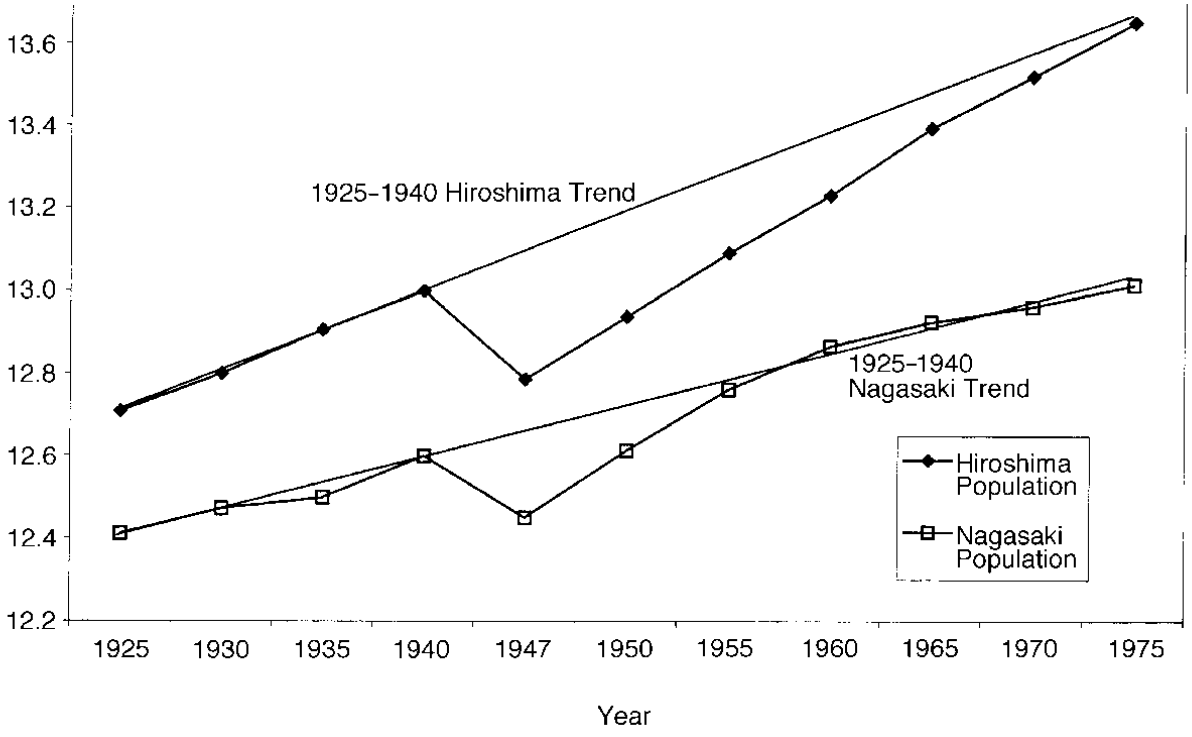



FIGURE 1. EFFECTS OF BOMBING ON CITIES WITH MORE THAN 30,000 INHABITANTS

Davis and Weinstein (2002), Figure 1.

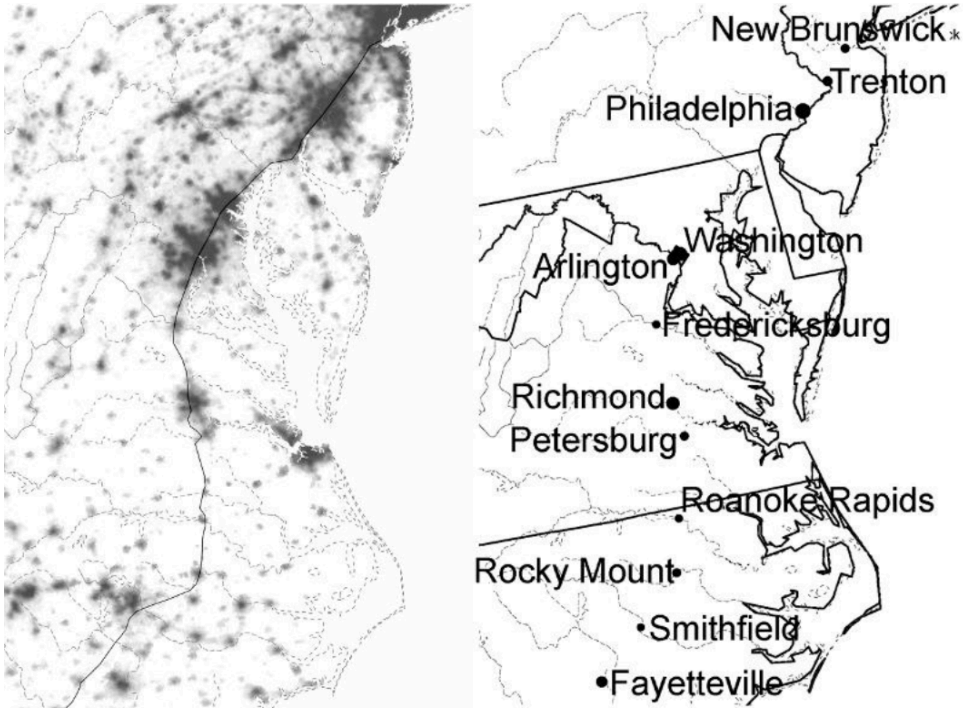
- Each circle is a Japanese city of more than 30,000 people with positive casualty rates.
- Across, growth from 1940 to 1947, the war years. Down the side, growth from 1947 to 1960.
- The cities with the largest wartime losses, on the left, had the fastest postwar growth.

# Japanese cities returned to their prewar growth paths within 15 years




- Allied bombing in 1944–1945 destroyed buildings and killed people across 303 Japanese cities in amounts that varied widely from city to city.
- Davis and Weinstein (2002) regress a city's growth from 1947 to 1960 on its loss from 1940 to 1947, instrumenting the loss with casualties and buildings destroyed per capita.
- The coefficient is close to $-1$. A city that lost a third of its people regained the whole third relative to other cities.
- The figure plots log population against each city's 1925–1940 trend. Nagasaki is back on its trend line by 1960 and Hiroshima by 1975.

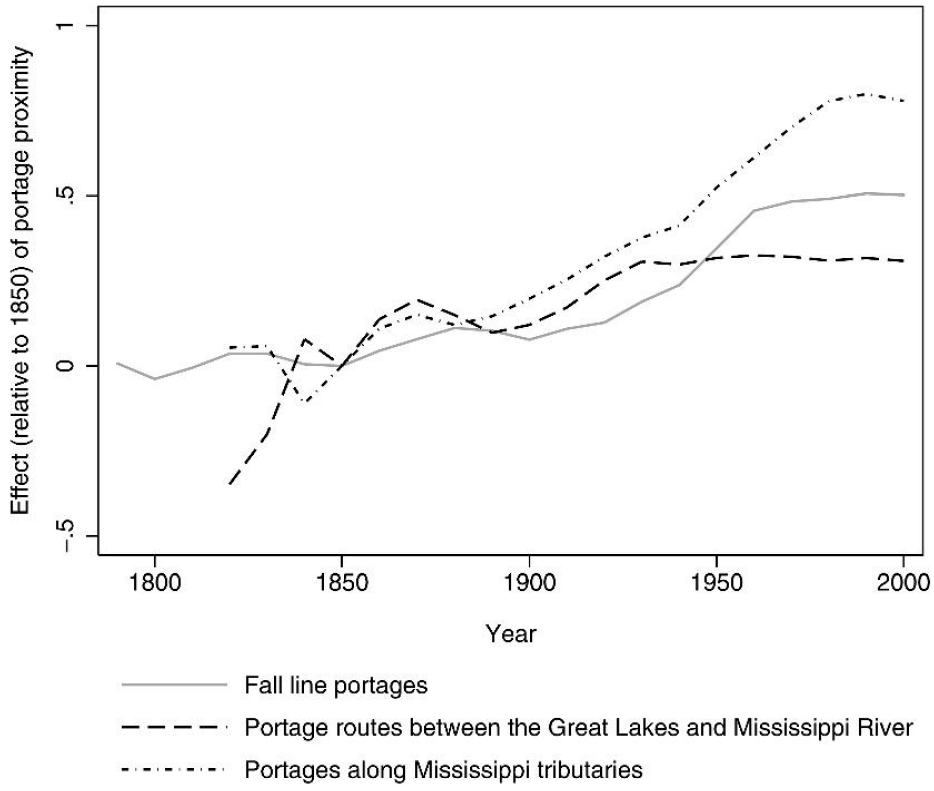
# The fall line runs through the East Coast's largest cities




Bleakley and Lin (2012), Figure IV.

- Left, night lights in 2003. The fall line is the solid black line and major rivers are dashed.
- Right, the cities on the fall line from North Carolina to New Jersey.
- Washington, Richmond, and Philadelphia sit where rivers cross the fall line.

# Portage sites kept their people after portage stopped mattering




Bleakley and Lin (2012), Figure V.

- Railroads made portage worthless by the late nineteenth century.
- Each line is the effect of portage proximity on log population density, by decade and relative to 1850, within watersheds. Solid gray is the fall line.
- The effect is larger today than in 1850, when portaging still had direct value.

# The standard errors of persistence

# An uncomfortable fact about this literature

> *"All of these objections crumble into irrelevance in the face of one blunt fact — the unusual explanatory power of these persistence variables. While a judicious choice of variables or time periods might coax a t statistic past 1.96, there would appear to be no way that the t statistics of four, five, or even larger that appear routinely in this literature could be the result of massaging regressions."*

- Conley and Kelly grant the point, then turn it around. If the t statistics are too large to have been manufactured, perhaps they are too large to be **real**.

- Most of these papers report p-values on the order of **0.001 or smaller**.

Conley and Kelly (2025), "The Standard Errors of Persistence."

# Persistence regressions are spatial regressions

- Both sides of these regressions are **maps**. Settler mortality is a map. Income today is a map. The mita boundary is a map.
- Maps are smooth. Neighboring places resemble each other, on the outcome and on the treatment alike.
- So the effective number of independent observations is far smaller than the number of rows in the data set, and a standard error computed as though the rows were independent is **far too small**.

# Two tests and a repair

- **The placebo test.** Replace the real treatment with spatial noise — a variable with the same smoothness and no possible relationship to the outcome. Run the study's own regression and inference. How often does pure noise come out significant?

- **The synthetic outcomes test.** Generate outcomes from a trend plus spatial noise, independent of treatment by construction and ask whether the real coefficient stands out against that reference distribution.

- **The repair.** Add a low-dimensional spatial basis to soak up the trend then use large-cluster standard errors. Their R package `spatInfer` does it.

# What their procedure actually does

1. **Build a spatial basis.** Lay a tensor spline over the map — think of four undulating threads running north to south crossed with four running east to west — and take the first few **principal components** of those terms, with the number chosen by BIC.

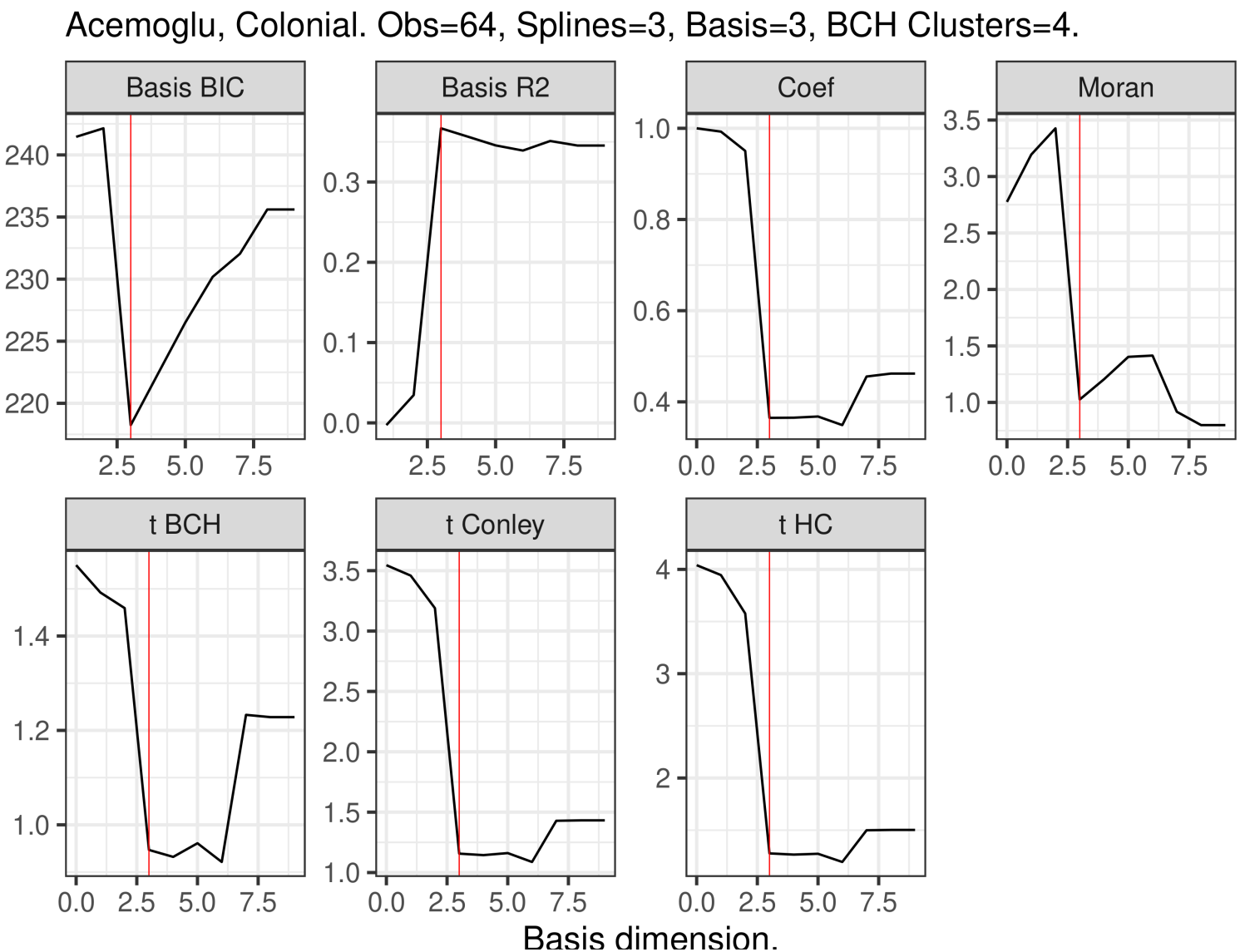
2. **Add them to the study's own regression.** They soak up the smooth spatial trend, which is what makes two unrelated maps correlate.

3. **Fix the standard errors.** Use a small number of large clusters, chosen by k-medoids, rather than many small ones. Neighboring small clusters are correlated with each other, so many small clusters understate uncertainty.

4. **Then test.** The placebo test replaces the treatment with spatial noise; the synthetic outcomes test replaces the outcome with trend plus spatial noise.

Note that step one is Dell's Panel A, generalized. She controlled for smooth geography with a cubic in latitude and longitude and lost significance. Conley and Kelly do the same thing more flexibly and to thirty papers at once.

# What it does to Acemoglu, Johnson and Robinson




Conley and Kelly (2025), Figure 6, applied to the first stage we looked at earlier. Moran is a z statistic for spatial correlation in the residuals.

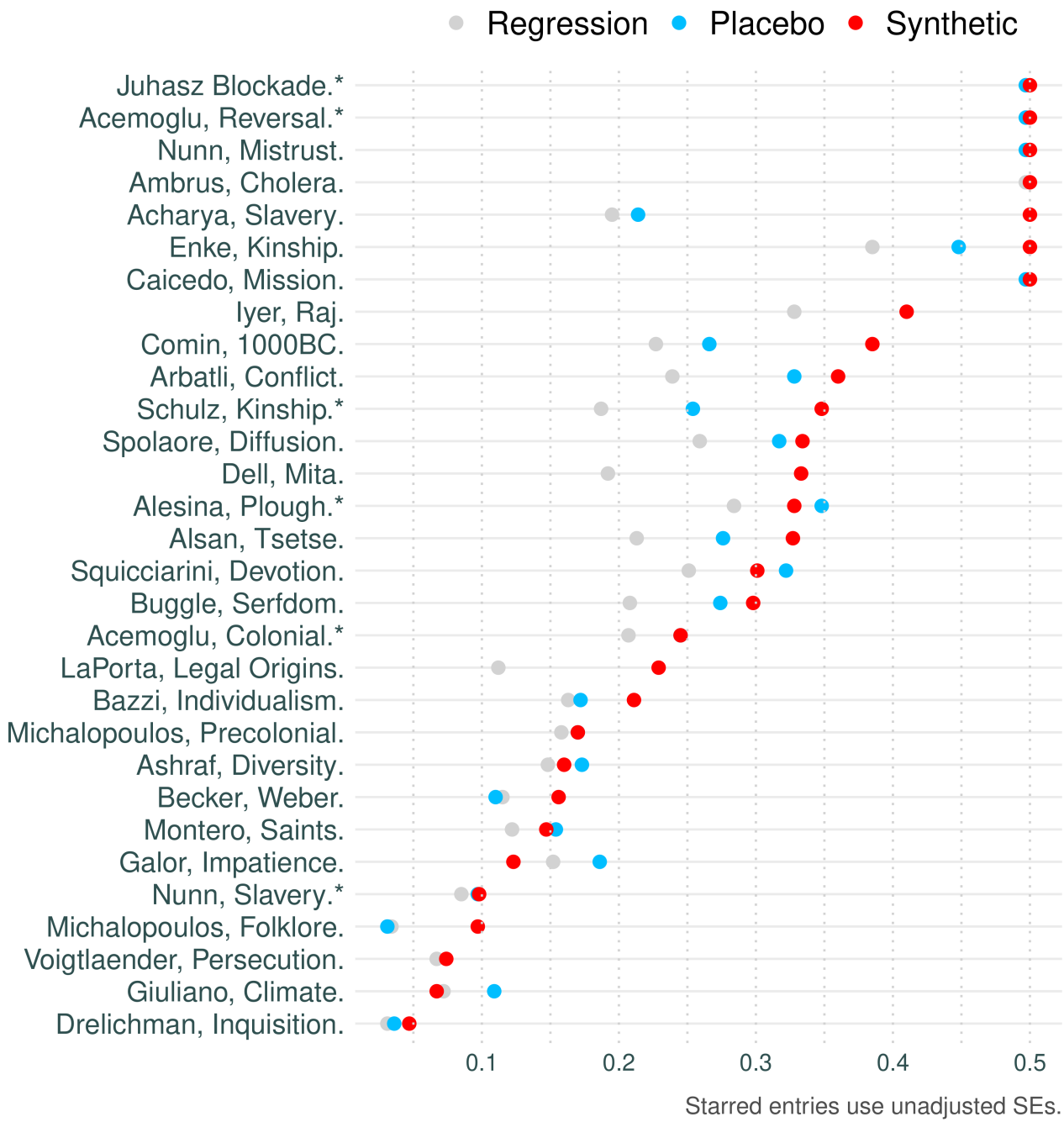
# Reading those seven panels

Every panel shares one horizontal axis, the **basis dimension**, which is how much smooth spatial structure has been taken out. The red line is where BIC says to stop, here at three.

| Panel | What it is | What it does here |
|---|---|---|
| Basis BIC | fit criterion for the basis | bottoms out at 3, which sets the red line |
| Basis R2 | what the basis alone explains | 0 to about 0.35 |
| Coef | AJR's first-stage slope | 1.00 down to about 0.36 |
| Moran | z statistic for spatial correlation left in the residuals | 3.4 down to about 1.0 |
| t BCH, t Conley, t HC | three ways of computing the standard error | 1.6, 3.5 and 4.1, all near 1 by 3 |

Look at the three t statistics at basis dimension **zero**, before anything is added. HC says 4.1, Conley 3.5, BCH 1.6. Same coefficient, same data, and the choice of standard error alone carries you from decisive to insignificant.

Then pause on Basis R2. Three principal components of a $3\times3$ spline, carrying no economics whatever, account for 35 percent of the variation in institutions across these 64 countries. That is Conley and Kelly's own figure.




Conley and Kelly (2025), Figure 7. Grey dots are each study's own regression re-run with the spatial basis. Starred rows use HC inference rather than clustered.

# Reading the audit

- One row per study, thirty of them, ordered by significance. The horizontal axis is a **p-value**, so left is stronger and the eye should travel to the bottom left.

- Three dots per row, all of them p-values against the **same null of no treatment effect**, reached three different ways.

  **Grey, Regression.** The study's own regression, re-run with the spatial basis and large-cluster standard errors.
  **Blue, Placebo.** The treatment replaced by spatial noise of the same smoothness. Left blank for binary treatments, which is why Dell has no blue dot.
  **Red, Synthetic.** The outcome replaced by trend plus spatial noise.

- Starred rows use HC standard errors, the rest use BCH.

A study survives if its grey dot sits left of **0.05**. Two of the thirty do.

The grey dot asks whether the effect survives. The blue and red ask the harder question, whether noise carrying the same spatial structure would have produced the finding anyway.

# Why the first stage is the worse place to lose

The first stage is not the result, so who cares if it wobbles?

- **Because everything is built on it.** If settler mortality cannot be distinguished from zero as a predictor of institutions once smooth geography is taken out, the instrument is **weak**.

- A weak instrument gives you an estimate whose bias runs back toward OLS and whose standard error cannot be trusted. The 2SLS number stops being informative about $\alpha$ **whatever the second stage looks like**.

Losing the second stage would say the effect is imprecise. Losing the first stage says the design is not identifying the thing it claimed to identify.

# What this does and does not mean

*"The fact that a treatment is insignificant at conventional levels does not necessarily mean that an effect is absent. It may simply not be possible to tie its magnitude down with any precision given the heavily trending and autocorrelated observations that are available. The legacies of history may be as overwhelming as their proponents insist, or as trivial as most historians suspect — we just cannot tell with the data at hand."*

- This is a claim about **what these data can establish**. Colonial history may have had enormous lasting effects, and these designs would still not show them.

- They re-run **one** regression per paper and insist they are not validating or disproving anybody.

- There is a second finding worth as much as the first. In two thirds of the studies at least one region's estimate has the **opposite sign** from the full-sample estimate. In one, every region does.

# Four steps before a spatial t statistic goes in a table

1. **Map the residuals.** If they come in patches the independent standard error is wrong. Report Moran's I with a stated neighborhood.

2. **Add a trend.** A quadratic in the coordinates for a regional study, region dummies for a global one. Report the coefficient with and without it. If the trend takes the coefficient the design cannot separate the treatment from a smooth spatial gradient.

3. **Run the placebo.** Draw noise with the treatment's spatial structure, rerun the regression a few hundred times, and report the share that clears your critical value. That share is the size of your test.

4. **Report more than one standard error.** HC, Conley at a stated distance, and a few large clusters with the right degrees of freedom.

Claude writes each of these in minutes. Choosing the neighborhood, the trend, the noise range, and the clusters is the part that requires knowing the setting.

# Running it with Claude (optional)

# Starting the project in Claude Code

Copy `examples/persistence_project/` to `~/wk09-demo`, open a terminal there, and start `claude`. The folder holds the Week 2 data and a `CLAUDE.md` that specifies the construction in words. It has no scripts.

## The request

```
Read CLAUDE.md. Write scripts/01_spatial_noise.R to do the whole construction and
run it. Report the coefficient on km_mainz and its five t statistics with and
without the quadratic trend, then the placebo summary: the share of draws with |t|
above 1.96, 3, and 4 under HC1, the largest |t|, the 95th percentile, the share of
draws with |t| at least as large as the real |t|, and the table of rejection
shares under each standard error with and without the trend.
```

Read the script before the numbers. Find the line that builds the noise covariance and check that the outcome is never touched. Find the critical values for the cluster columns.

# Expected output

These numbers come from the same construction, run when the slides were built. The placebo shares move by a few points from one seed to another. The five t statistics on the real regression should match exactly.

| | HC1 | Conley | country | 6 clusters | 4 clusters |
|---|---|---|---|---|---|
| real t, no trend | -5.33 | -2.84 | -3.35 | -3.00 | -2.28 |
| real t, with trend | -0.34 | -0.25 | -0.22 | -0.36 | – |
| noise rejects, no trend | 0.25 | 0.13 | 0.15 | 0.09 | 0.06 |
| noise rejects, with trend | 0.16 | 0.14 | 0.14 | 0.08 | 0.10 |

| | |
|---|---|
| coefficient on km_mainz | -0.000138 |
| share of draws with $\lvert t\rvert > 3$, HC1 | 0.08 |
| share with $\lvert t\rvert > 4$ | 0.02 |
| largest $\lvert t\rvert$ | about 5.8 |
| 95th percentile of $\lvert t\rvert$ | about 3.1 |
| share at least the real $\lvert t\rvert$ | 0.004, so 2 draws in 500 |

If the real t statistics differ check the sample first. Look for 744 cities and 163 with a press.

# Break

Five minutes. The clinic starts after it.

# Paper clinic

# Week 10

- Week 10 opens the text arc. A corpus becomes a document-term matrix and preprocessing choices decide much of what the matrix says.

- Michalopoulos and Xue (2021) links the two arcs. Oral traditions become measures of culture, joined to ethnic homelands as in Week 8. It is one of the two studies that survive Conley and Kelly's audit.

- Exercise 2 comes back with feedback. Presentations are four weeks out.

# References


Bleakley, H. and Lin, J. (2012). Portage and path dependence. *The Quarterly Journal of Economics*, 127(2):587–644.

Davis, D. R. and Weinstein, D. E. (2002). Bones, Bombs, and Break Points: The Geography of Economic Activity. *American Economic Review*, 92(5):1269–1289.

Michalopoulos, S. and Xue, M. M. (2021). Folklore. *The Quarterly Journal of Economics*, 136(4):1993–2046.

# Exercise 2 debrief

# The build half

- Every leave-one-out curve is U-shaped with its minimum between 1.5 and 2.5. France and the United Kingdom choose 1.5, Germany, Italy, and Spain choose 2. A curve that falls all the way to the smallest power left the predicted city in its own training set.

- The three largest gaps between inverse distance and nearest point sit where recorded cities are sparse and disagree. For Italy they are Bolzano, Asiago, and Varallo, all Alpine. For the United Kingdom only five cities differ by more than 20 points, because almost every city is within 50 km of a recorded one.

- Nine of the ten circles around Africa’s largest cities hold cells at 63. Addis Ababa’s holds none. Cairo and Giza overlap and share most of their cells.

- Summing density without cell area overstates the population inside a circle by about 20 percent. Taking the mean of the lights instead of the sum gives average brightness per person, which is a different quantity.

# The five errors in the audit script

| line | the error | what it does |
|---|---|---|
| 97 | `fun = "mean"` for the lights | light per person becomes average cell brightness per person. The correct top five is Libya, Gabon, Algeria, South Africa, Tunisia. The script prints Swaziland, Gabon, Tunisia, Gambia, Lesotho. |
| 60 | `which.max` on a correct leave-one-out curve | picks power 0.5, error 17.6, against 2 at 15.3. 908 of 1,797 cities move by more than 5 points. |
| 94, 98 | density resampled to 1 km, then summed with no area | population 969 million against 788 million, and the ratio runs from 1.16 to 1.44 with latitude. |
| 115–119 | the brightest large city is the first of nine tied at 63 | the finding is whichever city sorted first. Of the 20, 18 are at 60 or more. |
| 77–83 | country means for countries with no recorded city | Norway ranks first on two recorded cities. Denmark, the Netherlands, and Luxembourg get a mean on none. |

# The checks that would have caught them

- **Mean for sum.** One country's total against a published figure, or the units of the per-person number. Mean digital number per person has no meaning.

- **which.max.** Look at the cross-validation table and ask whether the chosen power sits at its minimum. The script printed the table and never compared the chosen power with its minimum.

- **Density without area.** Egypt in 2000 was about 68 million. The grid with areas gives 70, the resampled sum 94.

- **A tie at the ceiling.** Print the top of the ranking with ties visible. Nine cities read 63 because 63 is the most the sensor can read.

- **Means on no data.** Print the count of recorded cities beside each country mean.

- A sixth issue outside the planted five is that the script interpolates on longitude and latitude without projecting. gstat uses great-circle distance for unprojected points, so the numbers match a projected run. Project before interpolating regardless.